Date

1. $\because p\nmid a \quad \therefore ax^2+bx+c\equiv 0 \pmod p \Leftrightarrow 4a^2x^2+4abx+4ac\equiv 0 \pmod p$

$\Leftrightarrow (2ax+b)^2\equiv b^2-4ac$

∴解数为 $1+\left(\frac{b^2-4ac}{p}\right)$

2. (1) $x^2-y^2\equiv a \Leftrightarrow x^2\equiv a+y^2$ 设

$\therefore N(x^2-y^2=a)=\sum_{y=0}^{p-1} N(x^2\equiv i^2+a)=\sum_{y=0}^{p-1}\left(1+\left(\frac{y^2+a}{p}\right)\right)$

(2) $x^2-y^2=(x+y)(x-y)\equiv a$ ~~若 $p\nmid a$，则~~ 设 $x+y=u \quad x-y=v$

则 $x=\frac{v+u}{2} \quad y=\frac{u-v}{2} \quad x^2-y^2\equiv uv\equiv a$

若 p 为奇数，则 2 模 p 可逆，即 $x^2-y^2\equiv a$ 解的个数与 $uv\equiv a$ 解的个数相同

若 $p\nmid a$，则 $uv\equiv a \Leftrightarrow u\equiv av^{-1} \quad (v\neq 0)$

故解的个数为 $p-1$

若 $p\mid a$，则 $u\equiv 0$ 或 $v\equiv 0$ 故解的个数为 $2p-1$

$\therefore N(x^2-y^2=a)=\begin{cases} p-1, & p\nmid a \\ 2p-1, & p\mid a \end{cases}$

(3) 由(1)(2)得 $\sum_{y=0}^{p-1}\left(\frac{y^2+a}{p}\right)=\begin{cases} -1 & p\nmid a \\ p-1 & p\mid a \end{cases}$

3. 反(1)反证法 设 $h(x)=(x^{p-1}-1)-(x-1)(x-2)\cdots(x-(p-1))$

不是零多项式，而 $\deg h\le p-2$ 但 $h(x)$ 有 $p-1$ 个根 $1,2,3\cdots p-1$，矛盾

故 $x^{p-1}\equiv (x-1)(x-2)\cdots(x-(p-1))$

(2) 上式中取 $x=0$，即得

(3) $\prod_{i=1}^{\frac{p-1}{2}} r_i\equiv\left(\prod_{k=1}^{\frac{p-1}{2}} k\right)^2$

将 $(p-1)!=\left(\prod_{k=1}^{(p-1)/2} k\right)\left(\prod_{k=(p+1)/2}^{p-1} k\right)\equiv\left(\prod_{k=1}^{(p-1)/2} k\right)^2(-1)^{\frac{p-1}{2}}\equiv -1 \pmod p$

$\therefore \prod_{i=1}^{(p-1)/2} r_i\equiv\left(\prod_{k=1}^{(p-1)/2} k\right)^2\equiv(-1)^{\frac{p+1}{2}} \pmod p$

4. $n^4-n^2+1\equiv 0 \pmod p \quad \therefore (n^2+1)^2\equiv 3n^2 \pmod p \quad \left(n+\frac{1}{n}\right)^2\equiv 3 \pmod p$

$n^6-n^4+n^2\equiv n^6+n^2-(n^2-1)\equiv n^6+1\equiv 0 \quad \therefore n^6\equiv(n^3)^2\equiv -1$

$\therefore \left(\frac{-1}{p}\right)=1 \quad \left(\frac{3}{p}\right)=1 \quad \therefore \left(\frac{-3}{p}\right)=1$

$\therefore p\equiv 1 \pmod{12}$

5. $\left(\frac{7}{p}\right)\left(\frac{p}{7}\right)=(-1)^{\frac{7-1}{2}\cdot\frac{p-1}{2}} \quad \therefore \left(\frac{7}{p}\right)=(-1)^{\frac{3(p-1)}{4}}\left(\frac{p}{7}\right)$

∴若 $p\equiv 1 \pmod 4$ $\left(\frac{7}{p}\right)=\left(\frac{p}{7}\right)$ 此时若 $\left(\frac{p}{7}\right)=1$ $p\equiv 1,2,4 \pmod 7$

若 $p\equiv 3 \pmod 4$ $\left(\frac{7}{p}\right)=-\left(\frac{p}{7}\right)$ 此时若 $\left(\frac{7}{p}\right)=1$ $p\equiv 3,5,6 \pmod 7$

综上 $\left(\frac{p}{7}\right)=1 \Leftrightarrow p\equiv 1,3,9,15,23,25 \pmod{28}$

(1) 设 x 满足 $a_k x^k+\cdots+a_0\equiv 0$

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取 $\eta = a_k$，将易验证 $a_k\alpha$ 满足方程 $x^k + a_{k-1}x^{k-1} + a_k a_{k-2}x^{k-2} + \cdots + a_0 a_k^{k-1} = 0$

(2) 设 α 是 n 次首一多项式的根，β 是 m 次的根

考虑 $\mathbb{Z}$ 模 V 由 $\{\alpha^i\beta^j\gamma^k,\ 0\le i\le n-1,\ 0\le j\le m-1,\ 0\le k\le 1\}$

对任意 $\alpha^i\beta^j\gamma^k$，验证 $\gamma\alpha^i\beta^j\gamma^k \in V$

若 $k=0$，显然

若 $k=1$，$\gamma\alpha^i\beta^j\gamma^k = \alpha^i\beta^j\gamma^{k+2} = -\alpha^{i+1}\beta^j\gamma - \alpha^i\beta^{j+1}$ ①

若 $i+1 \ge n$，则可通过极小多项式降次价，$j+1\ge m$ 同理，故

故 ① 两项属于 V

综上 $\gamma V \subseteq V$ 又 γ 为代数整数